Core Java Study Plan Day 4

Programming Questions

1. Take n numbers from the user and count how many are positive, negative, and zero. Also print the largest positive number and smallest negative number.

2. Take start and end values and print all prime numbers in the range. Also print the total number of primes and their sum.

3. Take a number and find its largest digit, smallest digit, sum of digits, and count of even and odd digits using loops and if-else.

4. Take a number and repeatedly find the sum of its digits until a single-digit number is obtained. Check whether the final digit is even or odd.

5. Take a number and calculate the factorial of each digit. Find their total sum and check whether the number is a Strong number.

6. Take a number and reverse it. Check whether it is a palindrome. If it is a palindrome, additionally check whether it is divisible by 3 and 5.

7. Take start and end values and print all numbers that are divisible by 3 but not by 5. Also print their count and sum.

8. Take a number and find all its factors. Count the factors and use the count to determine whether the number is prime or not.

9. Create a menu-driven program using switch that repeatedly allows the user to:

1. **Reverse Number**
2. **Sum of Digits**
3. **Check Prime**
4. **Check Palindrome**
5. **Check Armstrong**
6. **Exit**

Continue displaying the menu until the user selects Exit.

10. Print the following pattern using nested loops:

```
    1
   121
  12321
 1234321
123454321
 1234321
  12321
   121
    1
```

APTITUDE TEST – SET 4

Percentage, Profit & Loss, Ratio & Proportion | 20 Questions

1. In a company, 60% of the employees are technical staff. Of the technical staff, 25% work in the night shift. If 90 technical employees work in the night shift, how many employees does the company have?

A) 500 B) 600 C) 750 D) 900

2. A candidate's score in Mathematics is 20% higher than his score in Aptitude. If his total score in the two subjects is 264, what is his Aptitude score?

A) 110 B) 120 C) 130 D) 140

3. The number of employees in a company is increased by 15%. Later, 8% of the new employees leave. What is the net percentage change in the workforce?

A) 5.8% increase B) 6.2% increase C) 7% increase D) 8% increase

4. A student answers 75% of the questions correctly. If 15 questions are incorrect, how many questions are there in the test?

A) 45 B) 50 C) 60 D) 75

5. A company's operating cost is 30% of its revenue. If revenue increases by 20% and operating cost increases by 10%, what percentage of the new revenue is the new operating cost?

A) 24% B) 25% C) 26% D) 27.5%

6. The population of a town is 48,000. It increases by 12.5% and then decreases by 20%. What is the final population?

A) 42,000 B) 43,200 C) 44,000 D) 45,000

7. A quantity is increased by 40%. By what percentage of the increased quantity must it be reduced to obtain a value that is 12% greater than the original?

A) 15% B) 20% C) 25% D) 28%

8. A trader buys 20 notebooks at ₹50 each. He sells 12 notebooks at ₹60 each and the remaining 8 at ₹45 each. What is his overall profit percentage?

A) 4% B) 5% C) 6% D) 8%

9. An article is sold for ₹2,400 at a profit of 20%. If the selling price is reduced by ₹240, what will be the resulting profit percentage?

A) 6% B) 8% C) 10% D) 12%

10. A trader buys an article for ₹1,800 and spends ₹200 on repairs. He wants a profit of 25% on the total investment. At what price should he sell it?

A) ₹2,400 B) ₹2,450 C) ₹2,500 D) ₹2,550

11. A shopkeeper gives a 15% discount on the marked price and still earns 19% profit. If the cost price is ₹2,000, what is the marked price?

A) ₹2,600 B) ₹2,700 C) ₹2,800 D) ₹2,900

12. A trader sells an article at a 20% profit. If he had purchased it for 20% less and sold it for ₹400 less, his profit would have been 25%. Find the original cost price.

A) ₹1,500 B) ₹1,800 C) ₹2,000 D) ₹2,400

13. A dealer buys 5 articles for ₹4,000. He sells 3 of them at a profit of 20% each and the remaining 2 at a loss of 10% each. What is his total profit?

A) ₹240 B) ₹280 C) ₹320 D) ₹360

14. An article is sold at a 25% loss. If its selling price is increased by ₹500, the loss becomes 5%. What is the cost price?

A) ₹2,000 B) ₹2,500 C) ₹3,000 D) ₹3,500

15. The ratio of the numbers of students in Classes A and B is 2:3. If 12 students move from B to A, the ratio becomes 3:4. What was the total number of students initially?

A) 280 B) 350 C) 420 D) 480

16. A:B = 3:4 and B:C = 6:7. If C = 84, find A.

A) 96 B) 108 C) 120 D) 126

17. A and B invest money in a business in the ratio 2:5. After 4 months, A doubles his investment. If the total profit after one year is ₹26,000, what is A's share?

A) ₹9,600 B) ₹10,000 C) ₹10,400 D) ₹12,000

18. The ratio of two numbers is 5:8. If 15 is added to each, the ratio becomes 2:3. Find the larger number.

A) 96 B) 108 C) 120 D) 128

19. A container has milk and water in the ratio 4:1. If 10 litres of water are added, the ratio becomes 2:1. What was the original quantity of the mixture?

A) 20 L B) 25 L C) 30 L D) 40 L

20. The ratio of the speeds of A and B is 4:5. If A takes 15 minutes more than B to cover the same distance, how long does B take?

A) 45 minutes B) 60 minutes C) 75 minutes D) 90 minutes

Puzzles

1. You have a balance scale and nine identical-looking balls. One ball is heavier than the other eight. Find the heavier ball using the balance scale only twice.

2. Five people are standing in a line. A is taller than B, B is taller than C, D is taller than E, and C is taller than E. Without knowing their exact heights, determine the possible order or explain why a unique order cannot be determined.

3. You have a room with four walls. Each wall has one window. A person walks into the room and sees three windows, but no mirror, doorway reflection, photograph, or screen is involved. How is this possible?

4. A man lives on the 20th floor of a building. Every morning he takes the lift to the ground floor. When he returns, he takes the lift only to the 12th floor and walks the remaining eight floors, except on rainy days when he goes directly to the 20th floor. Why?

5. You have a large chocolate bar divided into 12 equal pieces in a 3 × 4 arrangement. What is the minimum number of straight cuts needed to separate all 12 pieces completely, assuming pieces can be stacked after a cut?